

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

TIMETABLE AND VENUE MANAGEMENT SYSTEM (TVMS)

1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the Timetable and Venue Management System (TVMS).

The system is designed to facilitate academic timetable creation and management, venue allocation, emergency session scheduling, venue navigation, and real-time communication within educational institutions. The system aims to improve classroom utilization, reduce venue conflicts, assist users in locating assigned venues, and provide timely notifications whenever schedule changes occur.

This document serves as a reference for developers, project supervisors, stakeholders, system analysts, testers, and future maintainers of the system.

1.2 Problem Statement

Many institutes and universities still rely on manual or disconnected methods to manage academic timetables and venue allocation. This makes it difficult to coordinate venue usage efficiently, especially when timetable changes or emergency classes occur.

Additionally, students, lecturers, and visitors often experience difficulties locating assigned venues because campuses usually consist of many buildings, lecture halls, laboratories, and classrooms spread across large areas.

The absence of a centralized system for timetable management, venue allocation, venue location, and real-time communication contributes to venue conflicts, inefficient resource utilization, delays in class scheduling, and difficulties in disseminating schedule updates.

Therefore, there is a need to develop a smart, centralized, and real-time Timetable and Venue Management System capable of managing academic schedules, venue allocation, venue navigation, and communication within educational institutions.

1.3 Scope of the System

The Timetable and Venue Management System (TVMS) is a mobile-based platform developed to support academic scheduling and venue management activities.

The system shall provide:

- Academic timetable creation and management
- Venue management and allocation
- Emergency session scheduling
- Venue availability monitoring
- Venue location and navigation assistance
- Real-time notifications and communication
- Conflict detection and prevention
- Reporting and analytics
- User management and access control

The system creates and manages academic timetables directly and does not rely on importing timetable files.

1.4 Definitions and Acronyms

TVMS – Timetable and Venue Management System

CR – Class Representative

Venue – Classroom, lecture hall, laboratory, office, or any learning facility

Emergency Session – An unscheduled academic session created outside the official timetable

GPS – Global Positioning System

API – Application Programming Interface

JWT – JSON Web Token

RBAC – Role-Based Access Control

WebSocket – Real-time communication protocol

2. OVERALL DESCRIPTION

2.1 Product Perspective

The system operates as a centralized timetable and venue management platform that coordinates scheduling activities, venue allocation, emergency sessions, venue navigation, and real-time communication.

The system consists of:

- Flutter Mobile Application
- Django REST Framework Backend
- PostgreSQL Database

- Django Channels WebSocket Services
- GPS and Mapping Services
- Push Notification Services

2.2 Core System Functions

- Academic Timetable Creation and Management
- Venue Management
- Venue Allocation
- Emergency Session Scheduling
- Venue Navigation
- Real-Time Notifications
- Conflict Detection and Prevention
- Reporting and Analytics
- User Management
- Authentication and Authorization

2.3 User Classes

System Administrator

Responsibilities:

- Manage all users
- Configure system settings
- Manage permissions
- Monitor system performance
- Access all reports

Academic Administrator

Responsibilities:

- Create and manage timetables
- Manage venues
- Assign lecturers
- Monitor venue utilization
- Generate reports

Lecturer

Responsibilities:

- View assigned timetable

- View assigned venues
- Create emergency sessions
- Confirm session commencement
- Cancel sessions
- Receive notifications

Class Representative (CR)

Responsibilities:

- View timetable information
- View venue allocations
- Create emergency session requests
- Receive notifications
- Communicate schedule changes to students

Student

Responsibilities:

- View timetable
- View venue locations
- Navigate to assigned venues
- Receive notifications
- View emergency session updates

3. FUNCTIONAL REQUIREMENTS

3.1 Academic Timetable Management

FR-1: The system shall allow administrators to create academic timetables.

FR-2: The system shall allow administrators to edit timetable entries.

FR-3: The system shall allow administrators to delete timetable entries.

FR-4: Each timetable entry shall contain:

- Course
- Lecturer
- Program
- Venue
- Date
- Start Time

- End Time

FR-5: The system shall validate timetable information before saving.

FR-6: The system shall prevent venue conflicts.

FR-7: The system shall prevent lecturer schedule conflicts.

FR-8: The system shall maintain timetable modification history.

FR-9: The system shall support timetable filtering by:

- Lecturer
- Course
- Venue
- Program
- Date Range

3.2 Venue Management

FR-10: The system shall allow administrators to create venue records.

FR-11: Each venue shall contain:

- Venue Code
- Venue Name
- Building Name
- Floor Number
- Capacity
- Venue Type
- Coordinates
- Resources
- Accessibility Features
- Status

FR-12: The system shall support venue status monitoring and shall display all institute venues.

FR-13: The system shall support the following venue states:

- FREE
- IN_USE
- EXPIRED
- BOOKED

FR-14: The system shall maintain venue status history.

3.3 Venue Allocation

FR-15: The system shall allocate venues based on:

- Capacity requirements
- Resource requirements
- Availability
- Conflict prevention

FR-16: The system shall prevent double booking of venues and shall prevent lecturer time overlap.

FR-17: The system shall automatically determine venue availability also based on

- Timetable sessions
- Emergency bookings
- Confirmed sessions

FR-18: The system shall suggest suitable venues during emergency scheduling based on.

- Capacity $\geq$ expected students
- Required resources
- No time conflict
- Optimal capacity match

3.4 Venue Navigation

FR-19: The system shall store geographical coordinates for each venue.

FR-20: Users shall view venue locations on a map.

FR-21: The system shall display venue building information.

FR-22: The system shall provide navigation assistance to assigned venues.

FR-23: The system shall display floor information for multi-story buildings.

FR-24: The system shall assist new students and visitors in locating venues.

3.5 Emergency Session Management

FR-25: Authorized users shall create emergency sessions.

FR-26: Emergency sessions shall include:

- Session Title
- Course
- Venue
- Expected students (derived from course)
- Start Time

- End Time
- Required Resources
- Comment

FR-27: The system shall validate venue availability and capacity.

FR-28: The system shall validate lecturer availability.

FR-29: The system shall prevent emergency scheduling conflicts.

FR-30: The system shall log all emergency session activities.

FR-31: Emergency sessions may be cancelled before or during execution.

3.6 Session Confirmation

FR-32: Lecturers shall confirm session commencement through the mobile application, also should allowed by the system to postpone the session with show specific time, so system should send an notification with options to accept the session as confirm or postpone at what time

FR-33: Confirmation shall occur within a configurable grace period (40), before Lecturer should receive notification before 2-hours about entrance to enable early confirmation notification to students via CR

FR-34: Upon confirmation:

- Venue status becomes IN_USE
- Confirmation timestamp is recorded

FR-35: If confirmation is not received within the grace period:

- Session becomes EXPIRED
- Venue becomes FREE
- Notifications are generated

3.7 CR Notification Chain

FR-36: When the lecturer confirms the session, CR shall receive notifications.

- Session confirmation
- Session cancellation
- Venue release alerts

FR-37: When lecturer cancels session, CR shall receive notification.

FR-38: CR shall forward/commit notifications to students.

FR-39: Students shall only receive emergency notifications after CR action (unless marked critical).

FR-40: System shall log notification chain for audit.

3.8 Notifications

FR-41: The system shall support:

- Push Notifications and SMS notification
- In-App Notifications
- Email Notifications

FR-42: Notifications shall be generated for:

- Timetable updates
- Emergency sessions
- Session cancellations
- Venue changes
- Session confirmations

FR-43: Users shall configure notification preferences.

FR-44: Notifications shall be delivered in real time whenever possible.

3.9 Reporting and Analytics

FR-45: The system shall generate venue utilization reports.

FR-46: Reports shall include:

- Venue Usage Percentage
- Peak Usage Hours
- Venue Occupancy Statistics
- Conflict Prevention Statistics

FR-47: Reports shall support filtering by:

- Date Range
- Venue
- Department
- Lecturer

FR-48: Reports shall be exportable to PDF and Excel formats.

FR-49: Administrative dashboards shall display visual analytics.

4. NON-FUNCTIONAL REQUIREMENTS

4.1 Performance

- Support at least 500 concurrent users

- Mobile screen response time ≤ 3 seconds
- Real-time updates ≤ 2 seconds
- Venue search ≤ 2 seconds
- Timetable retrieval ≤ 3 seconds

4.2 Security

- Authentication shall use JWT
- Passwords shall be stored using Django secure hashing mechanisms
- HTTPS shall be enforced
- Role-Based Access Control shall be implemented
- SQL Injection prevention shall be enforced
- XSS protection shall be enforced
- CSRF protection shall be enforced where applicable
- API rate limiting shall be implemented

4.3 Privacy

- Minimum personal data collection
- Secure storage of personal information
- Access logging for sensitive data
- Controlled access to user records

4.4 Reliability

- Minimum uptime of 99.5%
- Daily database backups
- Recovery time objective ≤ 15 minutes
- Graceful degradation if WebSocket services become unavailable

4.5 Scalability

The system shall support:

- 1,000 venues
- 10,000 students
- 5,000 lecturers
- Horizontal backend scaling

5. SYSTEM ARCHITECTURE

5.1 Presentation Layer

The Presentation Layer consists of a Flutter Mobile Application that provides user interfaces for Students, Lecturers, Class Representatives (CRs), Department Administrators, and System Administrators. The mobile application enables users to access timetables, venue information,

notifications, emergency session management, and venue navigation services through an intuitive and responsive interface.

5.2 API Layer

The API Layer is implemented using Frappe Framework REST APIs and custom whitelisted API endpoints. It facilitates secure communication between the Flutter mobile application and the backend services by handling authentication, data exchange, and system requests.

5.3 Business Logic Layer

The Business Logic Layer is implemented within the Frappe Framework using DocTypes, Server Scripts, Python Controllers, Validation Rules, Permission Management, and Workflow Automation. This layer enforces system rules including timetable management, venue allocation, emergency session handling, lecturer attendance confirmation, notification processing, and venue status management.

5.4 Real-Time Communication Layer

The Real-Time Communication Layer utilizes Frappe Realtime Events, WebSockets, and Push Notifications to provide instant updates regarding venue status changes, timetable modifications, emergency session announcements, lecturer confirmations, and other critical academic events.

5.5 Background Processing Layer

The Background Processing Layer utilizes Frappe Scheduler Events and Background Jobs to execute automated tasks such as venue status updates, lecturer reminder notifications, session expiration checks, automatic venue releases, and scheduled system maintenance processes.

5.6 Database Layer

The Database Layer uses MariaDB to store and manage application data including user information, academic timetables, venue records, notifications, emergency sessions, attendance confirmations, audit logs, and system configurations.

5.7 Location Services Layer

The Location Services Layer integrates GPS, Mapping Services, and Campus Navigation functionalities to assist users in locating venues, identifying building locations, and navigating to classrooms within the institution. This layer supports venue discovery and improves accessibility for new students and visitors.

6. FUTURE ENHANCEMENTS

- AI-Based Timetable Generation
- AI-Based Venue Recommendation
- Automatic Conflict Prediction
- Indoor Navigation Support
- ERP Integration

- Attendance Tracking Integration
- Advanced Analytics Dashboard
- Offline Mobile Synchronization